Sheet 46 Thu 4 Mar. 2027 *Absolute value and square functions: sign and variations*

1. Warm-up · Quick Calculate $|-5|$, $(-5)^2$ and $|2-7|$.
2. Core · Method Give the table of variations of $x \mapsto |x|$ and its sign.
3. Core + · Reasoning Compare $(-3)^2$ and $(-2)^2$ then 3^2 and 2^2. Is the square function increasing on $\mathbb{R}$?
4. Review · Method Two fair dice are rolled. Probability that the sum is 7?
5. Thread · Method Programme: choose a number; add 1; square it; subtract the square of the starting number. What is the result for 6? What is the general expression?

Sheet 47 Mon 8 Mar. 2027 *Reciprocal function: variations; relative position of $y = x$ and $y = x^2$*

1. Warm-up · Quick Calculate $\frac{1}{2}$, $\frac{1}{-4}$ and $\frac{1}{0.1}$.
2. Core · Method Compare $\frac{1}{3}$ and $\frac{1}{5}$; what can be said about the variations of the reciprocal function on $]0;+\infty[$?
3. Core + · True/false True or false? Justify. For every real number $x > 0$, $x^2 \geqslant x$.
4. Review · Quick What are the front view and the top view of a cylinder standing on its base?
5. Thread · Prove Show that the product of two even integers is a multiple of 4.

Sheet 48 Tue 9 Mar. 2027 *Comparing $f(a)$ and $f(b)$; optimisation problems*

1. Warm-up · Quick $f(x) = x^2$. Compare $f(-3)$ and $f(2)$.
2. Core · Method $f(x) = x^2$, $a = -1.5$ and $b = -2$. Compare $f(a)$ and $f(b)$ without calculating, using the variations.
3. Core + · Optimisation $A(x) = x(10-x)$ for $x \in [0;10]$ (area of a rectangle with perimeter 20). Show that $A(x) = 25 - (x-5)^2$ and deduce the maximum.
4. Review · Quick 18 pupils out of 24 have an average above 10. Give the relative frequency as a percentage.
5. Thread · MCQ The solution of $\frac{x}{3} + 1 = \frac{7}{2}$ is:

   A. $\frac{15}{2}$ **B.** $\frac{9}{2}$ **C.** $\frac{21}{2}$ **D.** $\frac{5}{6}$

Cross-tabulating two qualitative variables

Sheet 49 Thu 11 Mar. 2027 *Cross-tabulating two qualitative variables*

1. Warm-up · Quick Express $\frac{24}{60}$ as a percentage.
2. Core · Quick In a class of 30 pupils, 18 are girls. What is the relative frequency of girls? of boys?
3. Core + · Classify Classify each variable: eye colour; baccalauréat grade (AB, B, TB); height in cm.
4. Review · Problem A rectangle of perimeter 20 cm has length x. Express its area $A(x)$; calculate $A(3)$ and $A(5)$.
5. Thread · Quick Calculate the exact volume of a cylinder of radius 2 cm and height 5 cm.

Sheet 50 Mon 15 Mar. 2027 *Qualitative variables; two-way frequency table*

1. Warm-up · Quick Complete: $12 + \ldots = 30$ and $22 + \ldots = 40$.
2. Core · Table Total 40 pupils. Girls: 22; canteen users (C): 25; girls who use the canteen: 15. Complete the two-way table.

	C	Not C	Total
Girls	15	…	22
Boys	…	…	…
Total	25	…	40

3. Core + · True/false True or false? Justify. "Satisfaction level (low, medium, high)" is a nominal qualitative variable.
4. Review · Problem Discount A: $-30\,\%$ on 80 000 Ar. Discount B: $-20\,\%$ then $-10\,\%$. Which is the better deal?
5. Thread · Method We know that $AB^2 + AC^2 = BC^2$. What conclusion can be drawn? Which property is used?

Sheet 51 Tue 16 Mar. 2027 *Marginal frequencies*

1. Warm-up · Quick Express $\dfrac{25}{40}$ as a percentage.
2. Core · Method Using the table from sheet 50: calculate the marginal frequency of girls and that of canteen users.
3. Core + · Quick The marginal frequency of girls is $55\,\%$. What is that of boys?
4. Review · Quick Calculate $(2a - b)^2$ for $a = 3$ and $b = -2$.
5. Thread · Quick $A(-1;2)$ and $B(3;2)$. What can be said about the line (AB)? Give AB.

Sheet 52 Thu 18 Mar. 2027 *Conditional frequencies*

1. Warm-up · Quick Give $\dfrac{15}{22}$ as a percentage rounded to the nearest tenth.
2. Core · Method With the same table: among the girls, what is the frequency of canteen users? Among the boys?
3. Core + · Distinguish Distinguish between "the frequency of girls among canteen users" and "the frequency of canteen users among girls".
4. Review · Construction Draw $[AB]$ of length 6 cm, then the circles with centres A and B and radius 4 cm, which meet at C and D. What type of quadrilateral is $ACBD$?
5. Thread · Quick Is 91 prime?

Sheet 53 Mon 22 Mar. 2027 *Completing a two-way table*

1. Warm-up · Quick Calculate $35\,\%$ of 80.
2. Core · Method 200 customers: $60\,\%$ are women; $25\,\%$ of the women and $50\,\%$ of the men buy online. Complete the two-way table.
3. Core + · Check A table states "row total: $45 + 55 = 100$" but "column total: $60 + 30 = 90$". What can be concluded?
4. Review · True/false True or false? Justify. After three consecutive "heads", the probability of getting "tails" on the next toss is greater than $\dfrac{1}{2}$.
5. Thread · Quick ABC is right-angled at A, $BC = 10$ cm, $\widehat{ABC} = 30°$. Calculate AC.